

Tracking Ebola

An African mobile-phone carrier can help researchers predict the spread of Ebola for health support <http://dgit.in/PredictEbola>

**Detect date-rape drugs**

New nail polish changes colour when it is dipped into drinks spiked with date-rape drugs <http://dgit.in/UndercoverColors>

Solving the world's clean water problem... one sip at a time

The world is facing a severe lack of potable water. Let's look at some technologies that are daring to meet this challenge head on

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How is it possible that despite two-thirds of the world being covered in water, people still suffer from a shortage of drinking water? The answer lies in the oceans, which make up most of the world's water. If you've ever had the misfortune of ingesting a gulp of seawater, you'd understand exactly why it isn't suitable for human consumption. Desalination is just one of the issues perplexing developing countries, whose inhabitants – approximately one in nine people – lack access to safe drinking water.

Safe drinking water, as per World Health Organization (WHO) guidelines, is water used for domestic purposes, drinking, cooking and personal hygiene. It has microbial, chemical and physical characteristics that meet WHO guidelines or national standards on drinking water quality.

Access denied

However, the following figure is quite telling in regards to how many people have access to this elixir of life in its pure form: Every year, 3.4 million people die from water-borne diseases, such as intestinal diarrhoea caused by microbially-contaminated water supplies. Nearly 99 percent of these deaths take place in the developing world. Vulnerable groups are children under the age of five, pregnant women and HIV patients.

Improved drinking water sources such as public standpipes, bore holes, protected dug wells and protected springs



Still cribbing about not having a PS4?

are more than a kilometre of walking distance from homes, or 100 metres of climb in hilly areas. It's usually the women and young girls of low-income countries who must sometimes walk more than six hours to go fetch water from streams and ponds – more likely than not chockfull of water-borne diseases. That's precious time that could be better utilised in working to raise themselves out of poverty through education and being able to hold on to jobs. According to U.N. estimates, if the proportion of people without access to safe water and basic sanitation were halved, countries globally would save \$7.3 billion per year in health care costs.

Deficient or non-existent sanitation and sewage disposal facilities are major reasons for non-potable water. However, the severity and reasons differ from location to location, and it's not just the rural poor who guzzle unclean water.



Fetching water from unclean sources

First world problems... for real

In August 2014, a Do-Not-Drink advisory was issued by the Toledo, Ohio government after above-standard readings for microcystin were reported in its water. It causes liver problems, diarrhoea and vomiting. While this was a one-off situation, the crisis lasting all of two days sent residents in a tizzy over getting hold of clean drinking water, in a scene that can only be comparable to the Black Friday sales in the U.S. Even earlier in the year, Charleston, West Virginia residents had to



Your Algae Margarita, sir

deal with a chemical spill in the Elk River contaminating their water supply. Millions of tons of toxic heavy metals such as arsenic, selenium, boron, cadmium, mercury and lead are disposed off in similar ways by coal-powered plants in waterways. Exposure to these dangerous chemicals has been proven to lead to birth defects, cancer and even death.

Mumbai, where approximately 40 percent of the population lives in slums, faces the challenge of providing safe drinking water to residents of such illegal settlements since they're, well, illegal. However, a survey conducted by the Office of the Reg-